

PD-split wins wherever it fits; above 29K, H100 only

SCENARIO (PREFIX / DECODE)	PREFIX	DECODE	KV GB	CS-3 ALONE	PD-SPLIT	H100 ALONE	GPU-STREAMED	BEST	DECODE SHARE
Qwen-Turbo p50 (short / tiny)	680	11	0.10	82	21	61	28	PD	48%
ServeGen R1 p50 (short / long)	208	792	0.03	745	726	3,654	888	PD	100%
Long reasoning (short / very long)	208	10,000	0.03	9,147	9,128	46,105	11,172	PD	100%
R1 p90	2,100	3,400	0.31	3,462	3,273	16,020	10,290	PD	99%
Qwen-Max p90 (mid / short)	2,400	433	0.35	668	453	2,078	1,472	PD	92%
Mooncake conv p50	6,900	350	1.02	1,096	482	1,834	2,850	PD	77%
Mooncake toolagent p90	17,000	507	2.51	2,435	1,012	3,103	9,468	PD	64%
Long-doc classification (long / tiny)	27,000	20	3.98	2,872	734	796	1,238	PD	4%
Mooncake conv p90 (long / short)	27,000	597	3.98	3,730	1,592	4,212	17,411	PD	56%
Mooncake conv p99	85,000	1,100	12.5	no fit	no fit	12,075	97,363	H100	80%
Claude Code median	197,000	5,800	29.1	no fit	no fit	88,215	1,158,677	H100	94%

ms per request, derived from measured rates; link 1.2 Tbit/s unmeasured. PD-split decodes on the wafer, so it keeps the 29K ceiling.